

For a simulation/dashboard like yours, use publicly available **Open Source Intelligence (OSINT)** and **think-tank data**. Exact real-time classified locations aren't publicly available (and many are deliberately obscured or mobile), but there are solid unclassified/open sources with approximate coordinates for major missile bases, silos, launch facilities, and control centers.

Best Sources for Missile Base Data

1. NTI (Nuclear Threat Initiative) WMD Facilities Directory (Strongly recommended starting point)

- Covers global **nuclear, chemical, and missile facilities** based on credible open-source material.
- Country-by-country breakdowns (e.g., Iran, North Korea, Russia, China, US, etc.).
- Includes missile production, test sites, operating bases, and storage.
- Link: [nti.org/education-center/facilities/](https://www.nti.org/education-center/facilities/)

2. Federation of American Scientists (FAS) Nuclear Notebooks

- Highly authoritative series on nuclear forces, including basing for nuclear-capable missiles (ICBMs, SLBMs, etc.).
- Regular updates on silo fields, road-mobile bases, submarine bases, and bomber bases.
- US, Russia, China, etc., covered in detail.

3. Arms Control Association – Worldwide Ballistic Missile Inventories

- Lists countries with ballistic missiles, types, status, and ranges. Good for context on operators.

4. CSIS Missile Threat Project

- Excellent database and profiles for individual missile systems and associated bases/launchers (search for specific countries).

5. Other Useful OSINT Resources

- **Beyond Parallel (CSIS):** Detailed reports on North Korean undeclared missile operating bases with geolocated info.
- Wikipedia lists (e.g., Nike sites — mostly historical US; missile launch facilities) for historical/decommissioned context.
- Google Earth / satellite imagery analysis from OSINT researchers (many FAS, CSIS, or independent analysts publish coordinates).
- ClimateViewer SAM Sites map (more air defense, but overlaps with some missile-related sites).

Implementation Tips for Your Dashboard

- **Data Format:** Export or scrape coordinates (lat/long) into GeoJSON, CSV, or a simple JSON array. Your Mapbox setup can ingest this easily as a custom layer (similar to your Global Incidents OSINT markers).
- **Categorization:**
 - Nuclear-capable (e.g., ICBM silos, SSBN bases).
 - Conventional / non-nuclear (e.g., shorter-range tactical/operational missile bases in Iran, North Korea, etc.).
 - Differentiate with icons, colors, or toggles (like your existing Aviation/Maritime groups).
- **Dynamic Behavior:** Treat them like your asset markers — clickable for details, with glassmorphic tooltips showing name, type (e.g., "Minuteman III silo field"), operator, and estimated capability.

- **Simulation Realism:**

- Many road-mobile systems (Russia, China, Iran, NK) don't have fixed "bases" but garrison/operating areas. Use representative points.
- Add a disclaimer in your UI: "Simulated OSINT-derived positions for training/educational purposes. Locations are approximate/public domain."
- For "live" feel, add occasional status updates via your backend (e.g., "Alert Level" or simulated activity).

Challenges & Best Practices

- **Accuracy:** Data is often approximate or outdated. Cross-reference multiple sources. Mobile launchers and underground facilities are hard to pinpoint exactly.
- **Sensitivity:** Stick to public data only. Avoid anything that could be seen as targeting current ops. Your nuclear sim is already disabled — keep this layer optional/toggleable.
- **Maintenance:** These don't change rapidly, so a static seed dataset updated periodically works well (like your telemetry seeds).
- **Legal/UI:** Attribute sources (e.g., "Data from FAS/NTI") in tooltips or a credits panel.

Start with NTI and FAS for the US, Russia, and China (biggest fixed infrastructure), then expand to regional players. If you need help parsing specific country data into JSON or integrating as a Mapbox source/layer, share more details on your current data structure! This will fit perfectly with your holographic globe and OSINT feed.